

Analyses: Classification with KNN

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Description

k-nearest neighbor analyses of the feature vectors per character sequence (loaded from ~/results/features_combined.csv). The results are stored in ~/results/classifier/.

Load Packages

If the libraries are not installed yet, you need to install them using, for example, the command: `install.packages("ggplot2")`.

```
library(dplyr)
library(class)
library(gmodels)
library(caret)
```

Load Data

Load data table with estimated feature values per sequence.

```
# load feature value estimations
estimations.df <- read.csv("~/Github/UpperPalCombinatorics/results/features_combined.csv")
head(estimations.df)
```

```
##      subcorpus      sequence num.char huni.chars hrate.chars  ttr.chars
## 1 Aurignacien ||| ///// _ V      12  1.8962406   1.5295563  0.41666667
## 2 Aurignacien      vvvvv         5  0.0000000   0.5711903  0.20000000
## 3 Aurignacien      vvvv vvvv      9  0.5032583   0.9060047  0.22222222
## 4 Aurignacien vvvvvvvvvvvvvvv      14  0.0000000   0.6379550  0.07142857
## 5 Aurignacien      |||           3  0.0000000   0.4308271  0.33333333
## 6 Aurignacien      |||           4  0.0000000   0.5070802  0.25000000
##      rm.chars
## 1 0.6363636
## 2 1.0000000
## 3 0.7500000
## 4 1.0000000
## 5 1.0000000
## 6 1.0000000
```

Select subcorpora for binary classification.

```
# choose subcorpora to compare
unique(estimations.df$subcorpus)
```

```
## [1] "Aurignacien"          "Cuneiform (Uruk V)"    "Cuneiform (Uruk IV)"
## [4] "Cuneiform (Uruk III)" "TeDDi"
```

```
subcorpus1 <- "Cuneiform (Uruk III)"
subcorpus2 <- "Aurignacien"
selected <- c(subcorpus1, subcorpus2)
estimations.subset <- estimations.df[estimations.df$subcorpus %in% selected, ]
```

Remove NAs (whole row)

```
estimations.subset <- na.omit(estimations.subset)
```

Center and scale the data

```
# first column is subcorpus column, the others are the features
# be aware that this needs to be changed in case of other table format
estimations.scaled <- cbind(estimations.subset[, c("subcorpus", "sequence", "num.char")],
                           scale(estimations.subset[, c("huni.chars", "hrate.chars",
                                                         "ttr.chars", "rm.chars")]))
```

Create Training and Test Sets

```
# Generating seed
set.seed(1234)
# Randomly generating our training and test samples with a respective ratio of 2/3 and 1/3
datasample <- sample(2, nrow(estimations.scaled), replace = TRUE, prob = c(0.67, 0.33))

# Generate training set (select rows randomly by using datasample vector)
train <- estimations.scaled[datasample == 1, 1:ncol(estimations.scaled)]
# Generate test set
test <- estimations.scaled[datasample == 2, 1:ncol(estimations.scaled)]
```

Initialize data frame

```
knn.results <- data.frame(k = numeric(0), accuracy = numeric(0), precision = numeric(0),
                          recall = numeric(0), f1 = numeric(0))
```

Building knn classifier

```
# choose maximum number of neighbors n
n = 10
# run a loop over different numbers of neighbors up to n
for (k in 1:n){
  # knn estimation of labels
  predictions.knn <- knn(train = train[, c("huni.chars",
                                           "hrate.chars",
                                           "ttr.chars",
                                           "rm.chars")],
```

```

        test = test[, c("huni.chars",
                        "hrate.chars",
                        "ttr.chars",
                        "rm.chars")],
        cl = train[, "subcorpus"], k = k)

# model evaluation
# creating a dataframe from known (true) test labels
test.labels <- test[, "subcorpus"]
# combining predicted and known classes
class.comparison <- data.frame(predictions.knn, test.labels)
# giving appropriate column names
names(class.comparison) <- c("predicted", "observed")
# inspecting our results table
head(class.comparison)
# get confusion matrix
cm <- confusionMatrix(data = class.comparison$predicted,
                      reference = as.factor(class.comparison$observed))
print(cm)
# get precision, recall, and f1 from the output list of confusionMatrix()
accuracy <- cm$overall['Accuracy']
f1 <- cm[["byClass"]][["F1"]]
recall <- cm[["byClass"]][["Recall"]]
precision <- cm[["byClass"]][["Precision"]]

# prepare data frame with results
local.results <- data.frame(k, accuracy, precision, recall, f1, row.names = NULL)
local.results.rounded <- round(local.results, 2)
# print(local.results.rounded)
knn.results <- rbind(knn.results, local.results.rounded)
# give predictions alongside the other information in data frame
knn.predictions.df <- cbind(test, predictions.knn)
}

```

```

## Confusion Matrix and Statistics
##
##               Reference
## Prediction      Aurignacien Cuneiform (Uruk III)
##   Aurignacien              49                  13
##   Cuneiform (Uruk III)      11                 277
##
##               Accuracy : 0.9314
##               95% CI : (0.8997, 0.9556)
##   No Information Rate : 0.8286
##   P-Value [Acc > NIR] : 1.239e-08
##
##               Kappa : 0.7618
##
##   Mcnemar's Test P-Value : 0.8383
##
##               Sensitivity : 0.8167
##               Specificity : 0.9552
##   Pos Pred Value : 0.7903
##   Neg Pred Value : 0.9618

```

```

##           Prevalence : 0.1714
##           Detection Rate : 0.1400
##           Detection Prevalence : 0.1771
##           Balanced Accuracy : 0.8859
##
##           'Positive' Class : Aurignacien
##
## Confusion Matrix and Statistics
##
##               Reference
## Prediction      Aurignacien Cuneiform (Uruk III)
##   Aurignacien              46              14
##   Cuneiform (Uruk III)      14             276
##
##           Accuracy : 0.92
##           95% CI : (0.8865, 0.9462)
##           No Information Rate : 0.8286
##           P-Value [Acc > NIR] : 5.655e-07
##
##           Kappa : 0.7184
##
## Mcnemar's Test P-Value : 1
##
##           Sensitivity : 0.7667
##           Specificity : 0.9517
##           Pos Pred Value : 0.7667
##           Neg Pred Value : 0.9517
##           Prevalence : 0.1714
##           Detection Rate : 0.1314
##           Detection Prevalence : 0.1714
##           Balanced Accuracy : 0.8592
##
##           'Positive' Class : Aurignacien
##
## Confusion Matrix and Statistics
##
##               Reference
## Prediction      Aurignacien Cuneiform (Uruk III)
##   Aurignacien              47              13
##   Cuneiform (Uruk III)      13             277
##
##           Accuracy : 0.9257
##           95% CI : (0.893, 0.9509)
##           No Information Rate : 0.8286
##           P-Value [Acc > NIR] : 9.068e-08
##
##           Kappa : 0.7385
##
## Mcnemar's Test P-Value : 1
##
##           Sensitivity : 0.7833
##           Specificity : 0.9552
##           Pos Pred Value : 0.7833
##           Neg Pred Value : 0.9552

```

```

##           Prevalence : 0.1714
##           Detection Rate : 0.1343
##           Detection Prevalence : 0.1714
##           Balanced Accuracy : 0.8693
##
##           'Positive' Class : Aurignacien
##
## Confusion Matrix and Statistics
##
##               Reference
## Prediction      Aurignacien Cuneiform (Uruk III)
##   Aurignacien              47              13
##   Cuneiform (Uruk III)      13             277
##
##           Accuracy : 0.9257
##           95% CI : (0.893, 0.9509)
##           No Information Rate : 0.8286
##           P-Value [Acc > NIR] : 9.068e-08
##
##           Kappa : 0.7385
##
## Mcnemar's Test P-Value : 1
##
##           Sensitivity : 0.7833
##           Specificity : 0.9552
##           Pos Pred Value : 0.7833
##           Neg Pred Value : 0.9552
##           Prevalence : 0.1714
##           Detection Rate : 0.1343
##           Detection Prevalence : 0.1714
##           Balanced Accuracy : 0.8693
##
##           'Positive' Class : Aurignacien
##
## Confusion Matrix and Statistics
##
##               Reference
## Prediction      Aurignacien Cuneiform (Uruk III)
##   Aurignacien              46              11
##   Cuneiform (Uruk III)      14             279
##
##           Accuracy : 0.9286
##           95% CI : (0.8964, 0.9532)
##           No Information Rate : 0.8286
##           P-Value [Acc > NIR] : 3.422e-08
##
##           Kappa : 0.7435
##
## Mcnemar's Test P-Value : 0.6892
##
##           Sensitivity : 0.7667
##           Specificity : 0.9621
##           Pos Pred Value : 0.8070
##           Neg Pred Value : 0.9522

```

```

##           Prevalence : 0.1714
##           Detection Rate : 0.1314
##           Detection Prevalence : 0.1629
##           Balanced Accuracy : 0.8644
##
##           'Positive' Class : Aurignacien
##
## Confusion Matrix and Statistics
##
##               Reference
## Prediction      Aurignacien Cuneiform (Uruk III)
##   Aurignacien              45              11
##   Cuneiform (Uruk III)      15             279
##
##           Accuracy : 0.9257
##           95% CI : (0.893, 0.9509)
##           No Information Rate : 0.8286
##           P-Value [Acc > NIR] : 9.068e-08
##
##           Kappa : 0.7314
##
## Mcnemar's Test P-Value : 0.5563
##
##           Sensitivity : 0.7500
##           Specificity : 0.9621
##           Pos Pred Value : 0.8036
##           Neg Pred Value : 0.9490
##           Prevalence : 0.1714
##           Detection Rate : 0.1286
##           Detection Prevalence : 0.1600
##           Balanced Accuracy : 0.8560
##
##           'Positive' Class : Aurignacien
##
## Confusion Matrix and Statistics
##
##               Reference
## Prediction      Aurignacien Cuneiform (Uruk III)
##   Aurignacien              44              12
##   Cuneiform (Uruk III)      16             278
##
##           Accuracy : 0.92
##           95% CI : (0.8865, 0.9462)
##           No Information Rate : 0.8286
##           P-Value [Acc > NIR] : 5.655e-07
##
##           Kappa : 0.7107
##
## Mcnemar's Test P-Value : 0.5708
##
##           Sensitivity : 0.7333
##           Specificity : 0.9586
##           Pos Pred Value : 0.7857
##           Neg Pred Value : 0.9456

```

```

##           Prevalence : 0.1714
##           Detection Rate : 0.1257
##           Detection Prevalence : 0.1600
##           Balanced Accuracy : 0.8460
##
##           'Positive' Class : Aurignacien
##
## Confusion Matrix and Statistics
##
##               Reference
## Prediction      Aurignacien Cuneiform (Uruk III)
##   Aurignacien              45                11
##   Cuneiform (Uruk III)      15               279
##
##           Accuracy : 0.9257
##           95% CI : (0.893, 0.9509)
##           No Information Rate : 0.8286
##           P-Value [Acc > NIR] : 9.068e-08
##
##           Kappa : 0.7314
##
## Mcnemar's Test P-Value : 0.5563
##
##           Sensitivity : 0.7500
##           Specificity : 0.9621
##           Pos Pred Value : 0.8036
##           Neg Pred Value : 0.9490
##           Prevalence : 0.1714
##           Detection Rate : 0.1286
##           Detection Prevalence : 0.1600
##           Balanced Accuracy : 0.8560
##
##           'Positive' Class : Aurignacien
##
## Confusion Matrix and Statistics
##
##               Reference
## Prediction      Aurignacien Cuneiform (Uruk III)
##   Aurignacien              47                10
##   Cuneiform (Uruk III)      13               280
##
##           Accuracy : 0.9343
##           95% CI : (0.903, 0.9579)
##           No Information Rate : 0.8286
##           P-Value [Acc > NIR] : 4.294e-09
##
##           Kappa : 0.764
##
## Mcnemar's Test P-Value : 0.6767
##
##           Sensitivity : 0.7833
##           Specificity : 0.9655
##           Pos Pred Value : 0.8246
##           Neg Pred Value : 0.9556

```

```

##           Prevalence : 0.1714
##           Detection Rate : 0.1343
##           Detection Prevalence : 0.1629
##           Balanced Accuracy : 0.8744
##
##           'Positive' Class : Aurignacien
##
## Confusion Matrix and Statistics
##
##               Reference
## Prediction      Aurignacien Cuneiform (Uruk III)
##   Aurignacien              45              10
##   Cuneiform (Uruk III)      15             280
##
##           Accuracy : 0.9286
##           95% CI : (0.8964, 0.9532)
##           No Information Rate : 0.8286
##           P-Value [Acc > NIR] : 3.422e-08
##
##           Kappa : 0.74
##
## Mcnemar's Test P-Value : 0.4237
##
##           Sensitivity : 0.7500
##           Specificity : 0.9655
##           Pos Pred Value : 0.8182
##           Neg Pred Value : 0.9492
##           Prevalence : 0.1714
##           Detection Rate : 0.1286
##           Detection Prevalence : 0.1571
##           Balanced Accuracy : 0.8578
##
##           'Positive' Class : Aurignacien
##

```

```
print(knn.results)
```

```

##      k accuracy precision recall  f1
## 1  1    0.93    0.79    0.82 0.80
## 2  2    0.92    0.77    0.77 0.77
## 3  3    0.93    0.78    0.78 0.78
## 4  4    0.93    0.78    0.78 0.78
## 5  5    0.93    0.81    0.77 0.79
## 6  6    0.93    0.80    0.75 0.78
## 7  7    0.92    0.79    0.73 0.76
## 8  8    0.93    0.80    0.75 0.78
## 9  9    0.93    0.82    0.78 0.80
## 10 10    0.93    0.82    0.75 0.78

```

Write to file.

```

# write results
write.csv(knn.results, file = paste(c("~/Github/UpperPalCombinatorics/results/classifier/KNN/results_KNN",
                                     subcorpus1, "_",
                                     subcorpus2, "_",
                                     ".csv")

```



```

                                ), collapse = ""),
                                row.names = F)

# write df with predictions for post-hoc analyses
# note that this writes the predictions only for the last run of the loop, i.e. the highest k
write.csv(knn.predictions.df, file = paste(c("~/Github/UpperPalCombinatorics/results/classifier/KNN/pre
                                subcorpus1, "_",
                                subcorpus2, "_",
                                ".csv"
                                ), collapse = ""),
                                row.names = F)

```